

# 大数据技术与基础

## 课后作业

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抓取一个公开网站上的数据，数据必须是多页，存为 csv 格式；如果数据是多个

pdf, 将其合并为一个 pdf。程序要尽量自动化, 减少人为干预。

代码:

```
import requests
import pandas as pd
import json
import time

pd.options.display.max_rows=500
headers = {'user-agent': 'Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:73.0)
Gecko/20100101 Firefox/73.0'}

url = 'https://c.m.163.com/ug/api/wuhan/app/data/list-total'    # 定义要访问的地址
r = requests.get(url, headers=headers)    # 使用 requests 发起请求

data_json = json.loads(r.text)
data = data_json['data']
data_province = data['areaTree'][2]['children']
areaTree = data['areaTree']

class spider_yiqing(object):

    # 将提取数据的方法封装成函数
    def get_data(data, info_list):
        info = pd.DataFrame(data)[info_list]    # 主要信息

        today_data = pd.DataFrame([i['today'] for i in data])    # 提取
today 的数据
        today_data.columns = ['today_' + i for i in today_data.columns]

        total_data = pd.DataFrame([i['total'] for i in data])
        total_data.columns = ['total_' + i for i in total_data.columns]

        return pd.concat([info, total_data, today_data], axis=1)

    def save_data(data,name):
        file_name
name+'_'+time.strftime('%Y_%m_%d',time.localtime(time.time()))+'.csv'
        data.to_csv(file_name,index=None,encoding='utf_8_sig')
        print(file_name+'保存成功! ')
```

```

if __name__ == '__main__':
    today_province = get_data(data_province, ['id', 'lastUpdateTime',
'name'])
    today_world = get_data(areaTree, ['id', 'lastUpdateTime', 'name'])
    save_data(today_province, 'today_province')
    save_data(today_world, 'today_world')

```

程序说明：

此脚本使用请求库向指定的 URL 发出 GET 请求。然后，它从响应中提取数据，并将其保存为 pandas DataFrame。提取的数据包括不同省份和全球新冠肺炎病例的信息。

运行截图：

```

In [1]: runfile('D:/新建文件夹 (4)/未命名3.py', wdir='D:/新建文件夹 (4)')
today_province_2023_10_16.csv保存成功！
today_world_2023_10_16.csv保存成功！

```

| 格式刷 粘贴 fx id |        |       |     |       |   |       |    |   |   |       |    |       |     |   |     |   |   |      |   |   |
|--------------|--------|-------|-----|-------|---|-------|----|---|---|-------|----|-------|-----|---|-----|---|---|------|---|---|
| A1           | fx id  |       |     |       |   |       |    |   |   |       |    |       |     |   |     |   |   |      |   |   |
| #            | A      | B     | C   | D     | E | F     | G  | H | I | J     | K  | L     | M   | N | O   | P | Q | R    | S | T |
| 7            | 220000 | ***** | 吉林  | 40485 | 0 | 40433 | 5  | 0 | 0 |       |    |       | 0   |   | 2   | 0 |   | -2   |   |   |
| 8            | 110000 | ***** | 北京  | 28860 | 0 | 17514 | 20 | 0 | 0 | 28266 | 11 | 16929 | 471 |   | 464 | 0 |   | 7    |   |   |
| 9            | 510000 | ***** | 四川  | 12023 | 0 | 10354 | 5  | 0 | 0 |       |    |       | 104 |   | 27  | 0 |   | 77   |   |   |
| 10           | 460000 | ***** | 海南  | 10436 | 0 | 9138  | 6  | 0 | 0 |       |    |       | 2   |   | 14  | 0 |   | -12  |   |   |
| 11           | 500000 | ***** | 重庆  | 10181 | 0 | 7480  | 7  | 0 | 0 |       |    |       | 209 |   | 45  | 0 |   | 164  |   |   |
| 12           | 410000 | ***** | 河南  | 9089  | 0 | 6880  | 23 | 0 | 0 |       |    |       | 42  |   | 325 | 0 |   | -283 |   |   |
| 13           | 150000 | ***** | 内蒙古 | 8394  | 0 | 8333  | 1  | 0 | 0 |       |    |       | 5   |   | 13  | 0 |   | -8   |   |   |
| 14           | 350000 | ***** | 福建  | 7651  | 0 | 6812  | 1  | 0 | 0 |       |    |       | 0   |   | 78  | 0 |   | -78  |   |   |
| 15           | 330000 | ***** | 浙江  | 6235  | 0 | 4851  | 1  | 0 | 0 |       |    |       | 37  |   | 20  | 0 |   | 17   |   |   |
| 16           | 610000 | ***** | 陕西  | 5589  | 0 | 5111  | 3  | 0 | 0 |       |    |       | 31  |   | 8   | 0 |   | 23   |   |   |
| 17           | 530000 | ***** | 云南  | 5561  | 0 | 4104  | 2  | 0 | 0 |       |    |       | 157 |   | 29  | 0 |   | 128  |   |   |
| 18           | 230000 | ***** | 黑龙江 | 5493  | 0 | 5253  | 13 | 0 | 0 |       |    |       | 16  |   | 11  | 0 |   | 5    |   |   |
| 19           | 140000 | ***** | 山西  | 5142  | 0 | 4339  | 0  | 0 | 0 |       |    |       | 46  |   | 295 | 0 |   | -249 |   |   |
| 20           | 370000 | ***** | 山东  | 4797  | 0 | 4518  | 8  | 0 | 0 |       |    |       | 25  |   | 104 | 0 |   | -79  |   |   |
| 21           | 320000 | ***** | 江苏  | 3808  | 0 | 3795  | 0  | 0 | 0 |       |    |       | 13  |   | 5   | 0 |   | 8    |   |   |
| 22           | 210000 | ***** | 辽宁  | 3684  | 0 | 2808  | 2  | 0 | 0 |       |    |       | 2   |   | 10  | 0 |   | -98  |   |   |

| fx 0 |         |            |       |            |           |            |           |           |            |            |           |            |           |           |           |             |   |   |   |  |
|------|---------|------------|-------|------------|-----------|------------|-----------|-----------|------------|------------|-----------|------------|-----------|-----------|-----------|-------------|---|---|---|--|
| H14  | fx 0    |            |       |            |           |            |           |           |            |            |           |            |           |           |           |             |   |   |   |  |
| #    | A       | B          | C     | D          | E         | F          | G         | H         | I          | J          | K         | L          | M         | N         | O         | P           | Q | R | S |  |
| 1    | id      | lastUpdate | name  | total_conf | total_sus | total_heal | total_dea | total_sev | total_inpu | today_conf | today_sus | today_heal | today_dea | today_sev | today_sto | today_input |   |   |   |  |
| 2    | 9577772 | *****      | 突尼斯   | 1147477    | 0         | 0          | 29279     | 0         | 0          | 0          |           | 0          | 0         |           |           |             |   |   |   |  |
| 3    | 9507896 | *****      | 塞内比亚  | 2440248    | 0         | 2410106    | 17498     | 0         | 0          | 518        |           | 743        | 5         |           |           |             |   |   |   |  |
| 4    | 0       | *****      | 中国    | 9506895    | 0         | 468612     | 31509     | 0         | 28824      | 46184      |           | 5552       | 78        |           | 24325     | 116         |   |   |   |  |
| 5    | 1       | *****      | 日本本土  | 1879839    | 0         | 1735909    | 18433     | 0         | 0          | 26881      |           | 8445       | 2         |           |           |             |   |   |   |  |
| 6    | 2       | *****      | 泰国    | 4718908    | 0         | 4649509    | 33505     | 0         | 0          | 0          |           | 0          | 0         |           |           |             |   |   |   |  |
| 7    | 3       | *****      | 新加坡   | 2196690    | 0         | 2122239    | 1710      | 0         | 0          | 710        |           | 1140       | 0         |           |           |             |   |   |   |  |
| 8    | 4       | *****      | 韩国    | 28659055   | 0         | 27451824   | 31790     | 0         | 0          | 58448      |           | 74644      | 46        |           |           |             |   |   |   |  |
| 9    | 5       | *****      | 澳大利亚  | 11077191   | 0         | 10869679   | 16940     | 0         | 0          | 2242       |           | 0          | 0         |           |           |             |   |   |   |  |
| 10   | 6       | *****      | 德国    | 37211937   | 0         | 36420100   | 160768    | 0         | 0          | 0          |           | 7300       | 0         |           |           |             |   |   |   |  |
| 11   | 7       | *****      | 美国    | 102242389  | 0         | 99225507   | 1116085   | 0         | 0          | 7187       |           | 51249      | 2         |           |           |             |   |   |   |  |
| 12   | 8       | *****      | 马来西亚  | 5023519    | 0         | 4972691    | 36831     | 0         | 0          | 609        |           | 736        | 0         |           |           |             |   |   |   |  |
| 13   | 9       | *****      | 越南    | 11524273   | 0         | 10610804   | 43184     | 0         | 0          | 71         |           | 41         | 0         |           |           |             |   |   |   |  |
| 14   | 89566   | *****      | 圣巴泰勒米 | 6          | 0         | 6          | 0         | 0         |            |            |           |            |           |           |           |             |   |   |   |  |
| 15   | 821     | *****      | 肯尼亚   | 342470     | 0         | 336521     | 5688      | 0         | 0          | 0          |           | 0          | 0         |           |           |             |   |   |   |  |
| 16   | 826     | *****      | 伊朗    | 7560792    | 0         | 7335938    | 144674    | 0         | 0          | 61         |           | 37         | 1         |           |           |             |   |   |   |  |
| 17   | 827     | *****      | 以色列   | 4756955    | 0         | 4733241    | 11988     | 0         | 0          | 831        |           | 941        | 0         |           |           |             |   |   |   |  |
| 18   | 828     | *****      | 毛利亚尼亚 | 16         | 0         | 6          | 2         | 0         |            |            |           |            |           |           |           |             |   |   |   |  |
| 19   | 829     | *****      | 黎巴嫩   | 1222055    | 0         | 1087587    | 10743     | 0         | 0          | 0          |           | 0          | 0         |           |           |             |   |   |   |  |
| 20   | 833     | *****      | 克罗地亚  | 1261936    | 0         | 1241600    | 17521     | 0         | 0          | 170        |           | 350        | 6         |           |           |             |   |   |   |  |
| 21   | 834     | *****      | 奥地利   | 5682776    | 0         | 5610355    | 21370     | 0         | 0          | 4679       |           | 5264       | 0         |           |           |             |   |   |   |  |
| 22   | 835     | *****      | 瑞士    | 4365574    | 0         | 4292193    | 14364     | 0         | 0          | 0          |           | 3398       | 0         |           |           |             |   |   |   |  |
| 23   | 839     | *****      | 希腊    | 5548487    | 0         | 5472780    | 34779     | 0         | 0          | 0          |           | 6822       | 0         |           |           |             |   |   |   |  |
| 24   | 95800   | *****      | 毛里求斯  | 41424      | 0         | 39694      | 1037      | 0         | 0          | 0          |           | 0          | 0         |           |           |             |   |   |   |  |
| 25   | 845     | *****      | 爱沙尼亚  | 611210     | 0         | 524990     | 2837      | 0         | 0          | 0          |           | 0          | 0         |           |           |             |   |   |   |  |
| 26   | 846     | *****      | 北马其顿  | 345402     | 0         | 335590     | 9608      | 0         | 0          | 0          |           | 0          | 0         |           |           |             |   |   |   |  |
| 27   | 850     | *****      | 白俄罗斯  | 994037     | 0         | 985592     | 7118      | 0         | 0          | 0          |           | 0          | 0         |           |           |             |   |   |   |  |
| 28   | 851     | *****      | 立陶宛   | 1286127    | 0         | 1268137    | 9473      | 0         | 0          | 430        |           | 71         | 1         |           |           |             |   |   |   |  |
| 29   | 852     | *****      | 阿塞拜疆  | 825965     | 0         | 815395     | 10006     | 0         | 0          | 72         |           | 72         | 0         |           |           |             |   |   |   |  |



葡萄酒数据集( wine.data) 搜集了法国不同产区葡萄酒的化学指标。试建立决策树、随机森林和 xgBoost3 种分类器模型，比较各种分类器在此数据集上的效果。

【提示】：每种分类器，需要对参数进行尝试，找出此种分类算法的较优模型，再与其他分类器性能进行比较。

代码：

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from xgboost import XGBClassifier

# 读取数据集
data = pd.read_csv('wine.data', header=None)

# 将数据集分为特征和标签
X = data.iloc[:, 1:]
y = data.iloc[:, 0]
from sklearn.preprocessing import LabelEncoder
#XGBoost 要求目标变量必须是从 0 开始的整数，而 y 变量取值是 [1, 2, 3],使用 LabelEncoder 进行转换
le = LabelEncoder()
xgy = le.fit_transform(y)

# 将数据集分为训练集和测试集
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3,
random_state=42)
X_train, X_test, xgy_train, xgy_test = train_test_split(X, xgy, test_size=0.3,
random_state=42)
# 建立决策树模型
dtc = DecisionTreeClassifier(random_state=42)
dtc.fit(X_train, y_train)
dtc_score = dtc.score(X_test, y_test)

# 建立随机森林模型
rfc = RandomForestClassifier(random_state=42)
rfc.fit(X_train, y_train)
rfc_score = rfc.score(X_test, y_test)

# 建立 xgBoost 模型

xgb = XGBClassifier(random_state=42)
```



```
xgb.fit(X_train, xgy_train)
xgb_score = xgb.score(X_test, xgy_test)
```

# 输出各个模型在测试集上的表现

```
print('Decision Tree Classifier score:', dtc_score)
print('Random Forest Classifier score:', rfc_score)
print('XGBoost Classifier score:', xgb_score)
```

程序说明:

从 wine.data 中读取数据将数据集中除第一列外的所有列赋值给变量 X, 将数据集中第一列赋值给变量 y。使用 LabelEncoder 进行对 y 的值进行转换, 将数据集分为训练集和测试集, test\_size 为 0.3, random\_state 为 42, 分别建立三种模型, 并对模型的性能进行测试, 输出各个模型在测试集上的表现, 发现随机森林的模型最好

运行截图

```
In [1]: runfile('D:/python/作业/新建文件夹/wine/5.py', wdir='D:/python/作业/新建文件夹/wine')
Decision Tree Classifier score: 0.9629629629629629
Random Forest Classifier score: 1.0
XGBoost Classifier score: 0.9629629629629629

This version of python seems to be incorrectly compiled
(internal generated filenames are not absolute).
This may make the debugger miss breakpoints.
Related bug: http://bugs.python.org/issue1666807

In [2]:
```